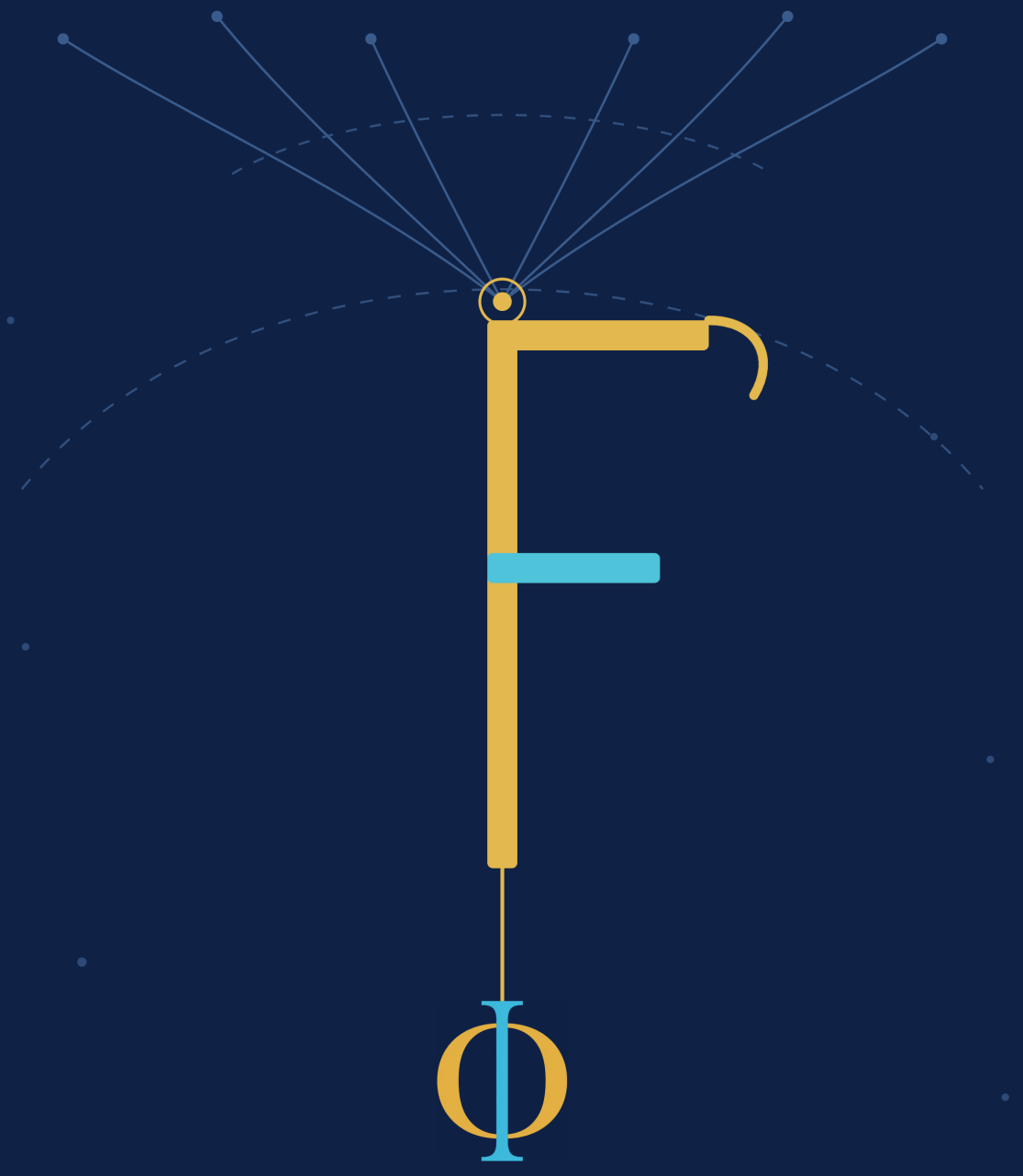
THINK LIKE Foλang



A developer's guide to elegant, idiomatic code

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Preface

Why This Book Exists

A programming language is more than syntax; it is the lens through which we view a system's architecture. It shapes how we define boundaries, express state, and whether complexity feels manageable or overwhelming.

FoLang was designed with that in mind.

The language reference gives you the normative definition — syntax, semantics, type rules, source structure, and capability boundaries. This book asks a different question:

> **How do we think when designing a solution in FoLang?**

Think Like FoLang is a guide to building the mental model that connects high-level architectural expression to low-level certainty. It is about looking at a problem and seeing how it wants to be represented.

The FoLang Mental Model

FoLang starts from a uniform premise:

> **Everything in FoLang is an object.**

That consistency means whether you are composing generic abstractions, leveraging pattern matching, or managing mutation, you are working within a single, cohesive model. Not every feature is forced into one implementation strategy — `co.lang.cstruct`, for instance, is an ABI-oriented value-semantic representation — but the reasoning stays unbroken from top to bottom.

This book will help you work out:

- When a type should carry domain meaning rather than just structural shape.
- Where behavior belongs, and which abstraction expresses the domain directly.
- How to use reference semantics, immutability, sharing, and copy-on-write cleanly.

- When pattern matching expresses intent better than branching.
- The boundary between value equality and reference identity.
- When an ordinary mechanism is enough, and when an advanced one is warranted.

Who This Book Is For

This book assumes you have written programs in at least one other language and are comfortable with types, functions, and objects. It does not assume you have used FoLang, and it does not assume a particular background language.

Where FoLang differs from what you may be used to, the book says so directly.

Specification and Book

Where exact syntax or semantics matter, the language reference governs. The examples here follow the current reference.

```
language-ref.md
    -> normative definition
    -> compiler-facing precision
    -> complete structural rules

Think Like FoLang
    -> developer-facing explanation
    -> mental models
    -> examples and idioms
    -> problem-solving
```

When a feature appears here, the goal is to explain why it exists, how to reason about it, and when it is appropriate. The reference remains the place to check the complete rule.

> **A note on stability.** FoLang is still evolving toward version 1.0. Syntax and semantics may be refined during that period, and examples in this book may be updated to match.

How This Book Is Organized

The book builds up in layers.

Part I — Learning to Think Like FoLang establishes the core model: objects, bindings, references, equality, and how expressions and matching fit together.

Part II — Building with FoLang develops the main abstractions: functions, units, structs, classes, types, generics, interfaces, signatures, type classes, extensions, and collections.

Part III — Designing FoLang Applications moves from single declarations to systems: packages, imports, project layout, applications, libraries, components, metadata, and execution models.

Part IV — Advanced FoLang covers mechanisms for cases where the ordinary abstractions are not enough: refinement and dependent types, macros, operators, reflection, specialization, native facilities, and longer case studies.

You do not need all of this before writing useful FoLang. The early chapters start with a small model and add to it when there is a reason to.

How to Read the Examples

FoLang uses explicit statement termination. A simple statement ends with `;`, while a block or declaration body ends with its closing `}`.

```
name co.lang.string = "Rao";
age  co.lang.int = 30;
```

```
(co.const.true).then({
    co.out.println("ready");
}).default({
    co.out.println("not ready");
});
```

Examples favor complete, readable forms first. Shorter forms and advanced facilities appear once their meaning is clear.

Where a FoLang construct resembles one from another language, the book uses FoLang's own vocabulary, since similar syntax does not always mean similar semantics.

What "Thinking Like FoLang" Means

Imagine FoLang itself looking at your system. What would it notice first? Which distinctions would it keep visible?

Consider a small example. Two Employee objects hold the same values:

```
a Employee = Employee{name: "Rao", id: 1};
b Employee = Employee{name: "Rao", id: 1};

a == b;             // true  — equal values
a.sameRef(b);       // false — different objects
```

Why does FoLang distinguish these two questions?

Imagine walking into a shop to buy a 100 g jar of coffee beans. On the shelf are ten sealed jars of the same product. They contain the same quantity and carry the same relevant product information.

From the point of view of those values, the jars are equal.

Now you pick up one jar and another customer picks up another.

Are you holding the **same jar**?

No.

The two jars may be equal in the relevant values we are comparing, but they are still two distinct physical objects.

FoLang thinks about managed objects in the same way:

```
equality
    -> do these objects have equal values?

identity
    -> do these references point to the same object?
```

That is why these are deliberately different questions:

```
a == b;            // true  — equal values
a.sameRef(b);      // false — different objects
```

If you open your jar and pour out half the beans, the other customer's jar remains full. The two jars were equal in value, but mutating one does not mutate the other because they are distinct objects.

Now bind a third name:

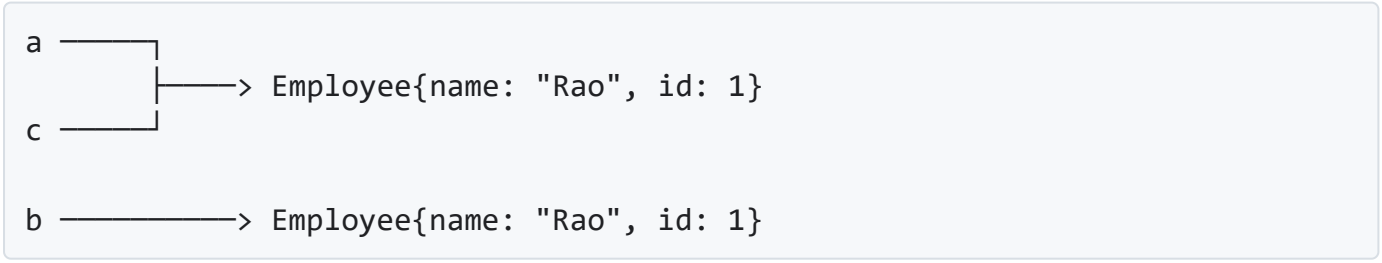
```
c := a;

c == a;            // true
c.sameRef(a);      // true  — c and a refer to the same object
```

Now imagine two people visiting a clock shop. They both notice a clock hanging on the wall. Each points to it and asks the salesperson about it. If we ask the salesperson whether both people are pointing/referring to the same clock, the answer is yes. There are two people making the reference, but both references point to one physical clock. FoLang thinks about `c := a` in the same way.

Here `c := a` does not create another Employee object. For managed objects, it binds `c` to the object already referenced by `a`.

If the shopkeeper adjusts the time on the clock, both people see the new time. They are not observing separate clocks; they are observing the state of the same physical clock.

Conceptually:



So `a`, `b`, and `c` can all be equal in value, while only `a` and `c` refer to the same object.

This distinction is central to thinking like FoLang:

- **equal** describes value;

- **the same object** describes identity;
- a binding is not the object itself.

The same habit applies elsewhere. `Employee{name: "Rao", id: 1}` is not only construction syntax; it explicitly states which type of object is being created. When you stop reading only the syntax and start aligning the domain model with the language's abstractions, you are beginning to think like FoLang.

```
Coffee jars
    -> two distinct objects
    -> initially equal values
    -> mutate one
    -> the other is unaffected
    -> sameRef(...) = false

People pointing to one clock
    -> two references
    -> one object
    -> mutate the clock
    -> both references observe the changed state
    -> sameRef(...) = true
```

A Developer's Guide to Elegance

Elegance here means clarity. It is not only about brevity; it is about leaving nothing to infer.

Elegant FoLang code makes the shape of the problem easy to see. A reader should be able to follow what the program is modelling, where its state lives, and why each abstraction is there. The way a component is written should reveal its intended semantics.

Clarity takes different forms. Sometimes it is a shorter expression. Sometimes it is a more explicit one. Often it is a well-chosen type that makes an entire category of question — and of bug — simply disappear.

The standard used throughout is a simple one:

> Does the code make the problem, the semantics, and the intent clearer?

When the answer is yes, the design is working.

Before We Begin

The first chapter begins with the question the rest of the book depends on:

> **How does FoLang think about a program?**

Once that model is in place, the syntax follows.